- **Title:** Problem / Problem continued...
- **Topic:** Hotel Count Occupied - Medium
- **Description:**
    - There are 10 floors in a hotel (numbered from 0 to 9). On each floor there are 26 rooms, each marked with a capital letter of the English alphabet (from 'A' to 'Z'). All rooms are initially free.
    - You are presented with the list of reservations, which consists of N four-character strings.
    - The first character of each string is + or -, which describes whether the room was booked or freed.
    - The second and third characters correspond to the number of the floor and letter of the room, respectively (e.g., 4C).
    - The fourth character indicates the type of the room (e.g., 'S' for Standard, 'D' for Deluxe, 'P' for Presidential). For example, +4CS means that a Standard room C on the 4th floor has just been booked, and -0GP means that a Presidential room G on the 0th floor has been freed.
    - The hotel has a maximum booking capacity for each specific room type across all floors. You will be provided with these capacities.

Rules for Processing Operations (Screenshot 2 & 3):

- **When a + (booking) operation is processed:**
    - If the specific room (e.g., room 4C) is already booked, the operation is ignored.
    - However, if the room is currently free, but booking it would cause the total number of booked rooms of its Type to exceed the Type's maximum capacity, then the booking attempt fails, and the room remains free.
    - If the room is free and booking it would not exceed its Type's capacity, the booking proceeds as usual.
- **When a - (freeing) operation is processed:**
    - If the specific room is already free, the operation is ignored.
    - Otherwise, the room is freed as usual, and the count of booked rooms for its Type decreases.

Task & Assumptions (Screenshot 3):

- Your task is to compute the number of rooms that are still booked after processing the entire list.
- You may assume that:
    - The list describes a correct sequence for freeing operations (only booked rooms are attempted to be freed).
    - All rooms are initially free.

    - The fourth character in the reservation string always correctly identifies the room's pre-assigned type.
    - Write a function that, given a Map (or Dictionary) roomTypeCapacities where keys are room type characters (e.g., 'S', 'D', 'P') and values are their integer capacities, and an array A consisting of N four-character strings representing the reservations list, returns an integer: the total number of rooms that are booked at the end (summing across all types).

Examples Breakdown (Screenshot 4):

Given:

- roomTypeCapacities = {'S': 2, 'D': 1, 'P': 1}
- reservationsList = ["+4CS", "+0AD", "+4CS", "+1BS", "+2CP", "+3DS", "-4CS", "+3DS", "-0AD", "-0AD"]

Step-by-step trace:

1. +4CS: Book Standard room C on 4th floor (S count: 1) $\rightarrow$ **Success**
2. +0AD: Book Deluxe room A on 0th floor (D count: 1) $\rightarrow$ **Success**
3. +4CS: Try to book 4C again $\rightarrow$ **Ignored** (already booked)
4. +1BS: Book Standard room B on 1st floor (S count: 2) $\rightarrow$ **Success**
5. +2CP: Book Presidential room C on 2nd floor (P count: 1) $\rightarrow$ **Success**
6. +3DS: Try to book Standard room D on 3rd floor $\rightarrow$ **Fails** (capacity for 'S' is 2, currently 2)
7. -4CS: Free Standard room C on 4th floor (S count: 1) $\rightarrow$ **Success**
8. +3DS: Book Standard room D on 3rd floor $\rightarrow$ **Success** (capacity available now, S count: 2)
9. -0AD: Free Deluxe room A on 0th floor (D count: 0) $\rightarrow$ **Success**
10. -0AD: Try to free Deluxe room A again $\rightarrow$ **Ignored** (already free)

Final result: **3** rooms remaining booked.